

Module 05 Assignment

Problem Solving with AI | Claude Code + JavaScript

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Batch	21st
Module	Module 05 - Problem Solving with AI
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Environment	VS Code Terminal

Q1. JavaScript Coding Practice

1. Convert Celsius to Fahrenheit

PROBLEM STATEMENT

Write a function that takes a temperature in Celsius and converts it to Fahrenheit using the formula: $F = (C \times 9/5) + 32$.

SOURCE CODE

```
function CelciusToFer(celcius){
  if(typeof celcius !== "number" || Number.isNaN(celcius)){
    return "The Value are Invalid!";
  } else{
    return (celcius*9/5)+32;
  }
}
console.log(CelciusToFer(44));
console.log(CelciusToFer(-22));
console.log(CelciusToFer("Riyad"));
```

INPUT AND OUTPUT

```
Input: 44      Output: 111.2
Input: -22     Output: -7.600000000000001
Input: Riyad   Output: The value is Invalid!
```

```
Problems Output Debug Console Terminal Ports
C:\Program Files\nodejs\node.exe .\assignment5.1.js
111.2
-7.600000000000001
The Value are Invalid!
assignment5.1.js:9
assignment5.1.js:10
assignment5.1.js:11
```

EXPLANATION

1. Checks whether the Celsius input is a valid number using `typeof` and `Number.isNaN()`. If invalid, it returns an error message.
2. If the input is valid, it converts Celsius to Fahrenheit using $(C \times 9/5) + 32$ and returns the result.

2. Find the Factorial of a Number

PROBLEM STATEMENT

Write a function that calculates the factorial of a given non-negative integer (e.g. $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$).

SOURCE CODE

```
function Factorial(n){
  if(typeof n !== "number" || !Number.isInteger(n) || n < 0){
    return "Invalid input: factorial needs a non-negative integer";
  }
  let result = 1;
  for(let i=2; i<=n; i++){
    result = result * i;
  }
  return result;
}
console.log(Factorial(7));
console.log(Factorial(15));
```

INPUT AND OUTPUT

```
Input: 7      Output: 5040
Input: 15     Output: 1307674368000
```

```
Problems Output Debug Console Terminal Ports
C:\Program Files\nodejs\node.exe .\assignment5.1.js
5040
1307674368000
assignment5.1.js:25
assignment5.1.js:26
```

EXPLANATION

3. Checks whether the input is a non-negative integer. If the input is invalid, it returns an error message.
4. Uses a for loop to multiply all integers from 2 to n and returns the final factorial result.

3. Check for Palindrome

PROBLEM STATEMENT

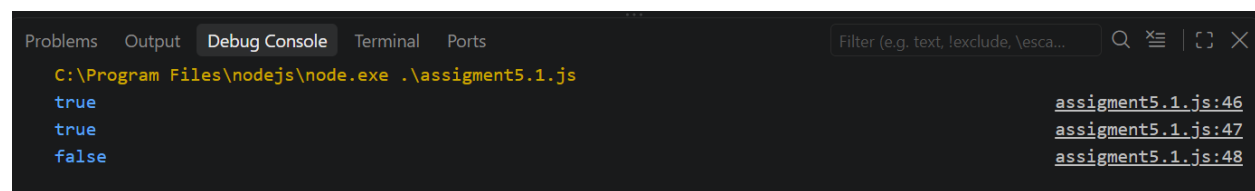
Write a function that checks if a given word (e.g. "racecar") reads the same backward as forward and returns true or false.

SOURCE CODE

```
function isPalindrome(word) {  
  
    if (typeof word !== "string" || word.length === 0) {  
        return "Invalid input: please provide a non-empty string";  
    }  
  
    let reversed = "";  
  
    for (let i = word.length - 1; i >= 0; i--) {  
        reversed = reversed + word[i];  
    }  
  
    return word === reversed;  
}  
  
console.log(isPalindrome("racecar"));  
console.log(isPalindrome("madam"));  
console.log(isPalindrome("riyad"));
```

INPUT AND OUTPUT

```
Input: racecar   Output: true  
Input: madam    Output: true  
Input: riyad    Output: false
```



The screenshot shows the VS Code interface with the 'Debug Console' tab active. The command prompt shows the execution of the script 'assignment5.1.js'. The output of the script is displayed on the right side of the console, showing 'true' for 'racecar', 'true' for 'madam', and 'false' for 'riyad'.

```
C:\Program Files\nodejs\node.exe .\assignment5.1.js  
true  
true  
false
```

assignment5.1.js:46
assignment5.1.js:47
assignment5.1.js:48

EXPLANATION

5. Checks whether the input is a non-empty string using `typeof` and `length`.
6. Reverses the string using a `for` loop and compares it with the original string to determine whether it is a palindrome.

4. Sum of Array Elements

PROBLEM STATEMENT

Write a function that takes an array of numbers and returns the sum of all elements in the array.

SOURCE CODE

```
function sumArray(numbers) {  
  if (!Array.isArray(numbers)) {  
    return "Invalid input: please provide an array";  
  }  
  let total = 0;  
  for (let i = 0; i < numbers.length; i++) {  
    if (typeof numbers[i] !== "number") {  
      return `Invalid element at index ${i}: not a number`;  
    }  
    total += numbers[i];  
  }  
  return total;  
}  
  
console.log(sumArray([1, 2, 3, 4, 5]));  
console.log(sumArray([10, 20, 30]));  
console.log(sumArray([]));
```

INPUT AND OUTPUT

Input: 1, 2, 3, 4, 5	Output: 2
Input: 10, 20, 30	Output: 3

The screenshot shows the VS Code interface with the 'Debug Console' tab active. The command prompt shows the execution of `node .\assignment5.1.js`. The output displays the results of the function calls: `15`, `60`, and `0`. On the right side, the corresponding line numbers in the source file are indicated: `assignment5.1.js:65`, `assignment5.1.js:66`, and `assignment5.1.js:67`.

EXPLANATION

7. Checks whether the input is an array and verifies that every element is a number.
8. Uses a for loop to add all array elements and returns the total sum.

5. FizzBuzz (Basic)

PROBLEM STATEMENT

Write a loop from 1 to 15 that prints "Fizz" for numbers divisible by 3, "Buzz" for numbers divisible by 5, "FizzBuzz" for numbers divisible by both, and the number itself otherwise.

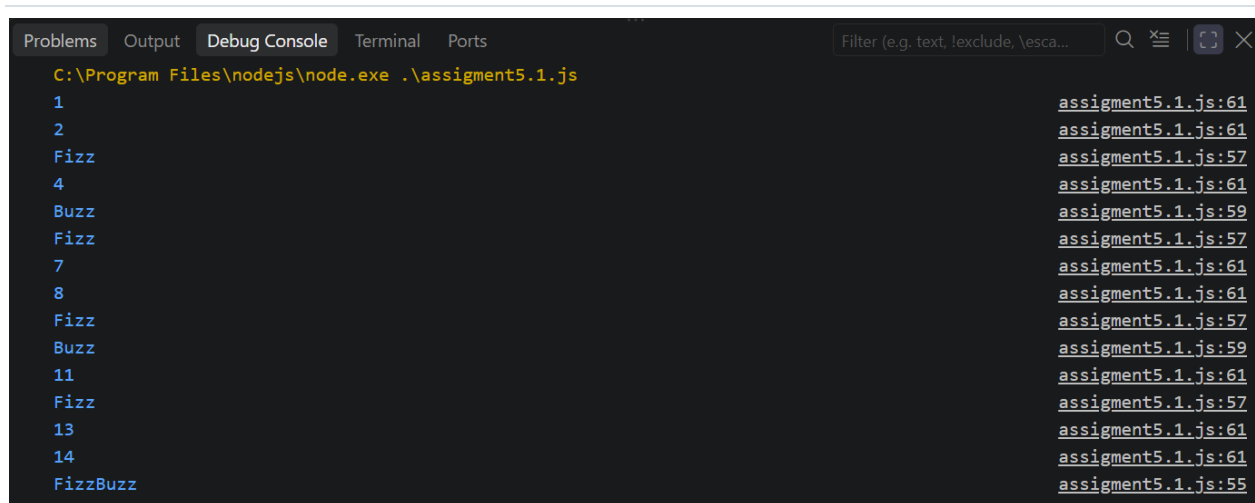
SOURCE CODE

```
function fizzBuzz(limit) {  
  for (let i = 1; i <= limit; i++) {  
    if (i % 15 === 0) {  
      console.log("FizzBuzz");  
    } else if (i % 3 === 0) {  
      console.log("Fizz");  
    } else if (i % 5 === 0) {  
      console.log("Buzz");  
    } else {  
      console.log(i);  
    }  
  }  
}  
  
fizzBuzz(15);
```

INPUT AND OUTPUT

Input: 15

Output: 1, 2, Fizz, 4, Buzz, Fizz, 7, 8, Fizz, Buzz, 11, Fizz, 13, 14 FizzBuzz



```
C:\Program Files\nodejs\node.exe .\assignment5.1.js
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
```

EXPLANATION

9. Outer Iterates from 1 to the given limit and checks each number using the % operator.
10. Prints "Fizz" for multiples of 3, "Buzz" for multiples of 5, and "FizzBuzz" for multiples of both.
11. console.log() to display each result immediately; return is not used because the function does not need to return a final value.

---The End---